

# AXONA

## *Synopsis and Implications*

David A. Smith · *Axona.net* · v0.1 · 2026-06-12

*During the mechanical ages we had extended our bodies in space. Today, after more than a century of electric technology, we have extended our central nervous system itself in a global embrace, abolishing both space and time as far as our planet is concerned.*

— Marshall McLuhan, *Understanding Media: The Extensions of Man*, 1964

### *What it is*

Axona is a peer-to-peer substrate for addressing, routing, and publish/subscribe messaging that runs with no platform in the middle. Three properties define it. It is **self-authenticating**: a participant's identity is its own cryptographic keypair, verified directly by its peers.<sup>1</sup> It is **geo-aware but not geo-fenced**: node addresses carry a coarse geographic prefix that biases routing toward locality — an area code, not a wall — so traffic stays efficient and community-scaled while no one is partitioned out. And it is **infrastructurally diffuse**: a single bootstrap broker introduces new peers, but the mesh sustains itself peer-to-peer, converges its state through replication and anti-entropy, and runs in an ordinary browser or an aging phone.<sup>2</sup>

We can harden the security model, close failure modes, and tune the protocol, and each such change nudges the network's behavior. But we cannot *measure* that behavior. We can measure latency. The social consequence travels on the same wire and is invisible to us.

<sup>1</sup> No certificate authority, no registrar, no identity broker to petition, surveil, or revoke through. You are your key.

<sup>2</sup> The entire cost of entry: a self-generated key and an open browser tab — no install, no account, no server.

### *What is genuinely new*

We have built censorship-resistant media before — onion routing, content-addressed storage, gossip social networks. Axona's pub/sub layer extends that lineage with a sharper consequence: it furnishes **ungovernable mobilization and un-censorable assembly**. A topic cannot be deplatformed, because there is no platform; a message cannot be reliably unsaid, because retraction is best-effort and any holder of the bytes keeps them; a movement cannot be switched off at a chokepoint, because the chokepoint does not exist. For people organizing under hostile infrastructure — dissidents, mutual-aid networks, civil-defense communications when centralized systems fail or are seized — this is a profound civic good. Assembly is among the

oldest of liberties, and Axona returns it to a form no intermediary can withdraw.

The qualitatively new element is *who is on the network*. Its participants are no longer only people: increasingly the publishers and subscribers are AI agents, acting beside us and on our behalf. McLuhan watched electric media extend the human nervous system across the planet; Axona is connective tissue for the next reach of that extension — a shared medium through which an augmented, human-and-machine intellect can sense, deliberate, and act as one body. Here human judgment and machine reach are not rivals but complements wired into a common fabric and amplifying one another: a person’s intent travels at the reach of the swarm, an agent’s findings surface to the people best placed to weigh them, and the loop closes with no intermediary standing between a thought and its expression.<sup>3</sup> Built well, this is a genuinely **cooperative and mutually empowering intelligence** — a planetary nervous system owned by its participants rather than rented from a platform.

### *The dual-use reality*

The properties that shelter the dissident shelter the harasser; the assembly that cannot be censored is also the assembly that cannot be moderated; and a medium in which machine-speed cascades run with no platform in the middle to dampen them carries a disinformation flood as readily as a flood warning.<sup>4</sup> This is not a flaw to be engineered away — it is the *same property* seen from two sides. **Uncensorable and un-governable are one fact.** To build a medium that no power can silence is to build a medium that no power can stop, including powers we would want stopped, and including non-human actors operating at a tempo our institutions were never designed to match.

### *Why we proceed, and what we owe*

We build it because the alternative — assembly and speech held permanently at the discretion of a few intermediaries, now augmented by AI that those same intermediaries control — is its own grave risk. A neutral, self-owned substrate is a hedge against that concentration. But neutrality is not innocence. The responsibility that platforms discharged badly does not vanish when the platform does; it devolves onto the architecture, onto the applications built atop it, and onto the norms of the communities — human and agentic — that inhabit it. Our obligation is to name this plainly, to shape the affordances that govern a cascade — the cost of an identity, the pull of proximity, the

<sup>3</sup> The “extensions of man” become extensions of *man-with-machine*; the substrate that carries them should belong to the people and agents it connects, not to a broker sitting between them.

<sup>4</sup> Markets already gave us the flash crash as a preview. Axona generalizes that dynamic from prices to arbitrary coordination — and removes the one thing every prior platform, however biased or captured, also happened to supply: a brake.

reach of a retraction, the rate of spread — with that cascade in mind,  
and to keep studying the process we have set loose.

*Axona is not a wiring diagram. Axona is a process. It is defined less  
by its topology than by what flows through it — and that is no longer  
on paper. **The first version is in the wild; anyone can use it.***